

RESEARCH ARTICLE

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Evaluation of a lifestyle intervention for women with gestational diabetes mellitus based on the medical internet of things: a randomized controlled trial with mid-sample verification

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ABSTRACT

Objective: To evaluate the effect of lifestyle intervention using a self-management model based on the medical Internet of Things (mIoT) for women with gestational diabetes mellitus (GDM).

Methods: 240 eligible participants were initially enrolled, with 225 completing the trial (116 in the intervention group, 109 in the control group) due to 15 dropouts (4 in the intervention, 11 in the control) caused by delivery at non-study hospitals. The control group regularly received the lifestyle interventions. The intervention group received the same interventions based on mIoT. Data on glucose metabolism and lipid metabolism were collected at 4 weeks and 8 weeks after the intervention and pre-delivery at the admission. Maternal and infant outcomes were compared between the two groups.

Results: After the intervention, the fast blood glucose (FBG) and weight gain in the intervention group were significantly decreased than those in the control group. The glucose-metabolism indexes and lipid metabolism indexes were significantly lower in the intervention group than the control group. The rates of preeclampsia, preterm birth, neonatal hypoglycemia, and fetal macrosomia were also significantly lower in the intervention group than those in the control group.

Conclusion: The lifestyle intervention based on the medical Internet of things could improve the compliance and effectiveness of self-management in women with gestational diabetes mellitus, and it had a good clinical effect on their metabolic status in terms of blood glucose and lipids in late pregnancy and pregnancy outcomes.

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Gestational diabetes mellitus; lifestyle intervention; medical internet of things; glucose metabolism; lipid metabolism

Introduction

Gestational diabetes mellitus (GDM) is defined as diabetes mellitus or glucose intolerance occurring for the first time in pregnancy (1). Although its different diagnostic criteria in different regions, GDM affects approximately 15%–20% of pregnancies (2), or as many as 30% in some parts of the world (3). GDM is an inflammatory disease that can destroy islet cells and aggravate insulin resistance (4). Plasma C-reactive protein (CRP), a sensitive indicator of chronic subclinical inflammation, is a predictor of insulin resistance syndrome independently associated with insulin sensitivity and metabolic disorders (5). Insulin resistance and glycolipid metabolism disorder due to insulin resistance are recognized as the physiological and pathological basis of GDM (5,6). They are closely related to adverse pregnancy outcomes, such as preeclampsia, premature delivery, large for gestational age, and increased cesarean section rate (7,8). Moreover, GDM is also associated with long-term negative maternal and fetal outcomes (9).

Current treatment programs mainly include changed lifestyle, more physical activities, and improved diet quality based on medical nutrition therapy (10,11). Medical nutrition therapy aims to achieve and maintain euglycaemia by consuming appropriate dietary portions, distributing carbohydrates and consuming foods with a lower glycemic index (10,12). Most GDM women can achieve good blood glucose control

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through lifestyle intervention (10,11). The effectiveness of lifestyle interventions depends on the frequency of medical nutrition therapy consultation and compliance to the intervention (10,12). If lifestyle interventions alone are not effective, insulin or other glucose-lowering medication is usually prescribed (13,14).

Orem's (1995) self-care model is used as the framework to provide a conceptual structure for the supportive educative program for GDM women (15). In Orem's (2001) model, nursing aims to help to meet their therapeutic self-care demands (16). These are defined as the sum of self-care actions needed to fulfil all of a person's self-care requirements (16). Nursing is required when there is a deficit between self-care needs and self-care abilities. Cradock et al. have previously reviewed those four important factors, including "instruction on how to perform a behavior," "action planning," "goal setting," and "feedback and monitoring," which were fundamental for the success of behavior modification techniques (17). Accordingly, an original system that utilizes wearable monitoring devices loaded with the Internet of Things (IoT) and facilitates telehealth coaching by public health nurses was developed to cover these four factors. These devices and mobile applications are now increasingly used and integrated with telemedicine and telehealth via the medical Internet of Things (mIoT). It is characterized by a defined architecture design, including hardware and software such as wearable devices, sensors, smartphones, medical applications, and medical station analyzers for further data collection, storage, analysis, diagnosis, and management (18,19). It has been reported that lifestyle intervention based on mIoT could improve the effectiveness of self-management and increase compliance with treatment in elderly patients with type 2 diabetes mellitus (T2DM) (20).

Therefore, we aimed to explore whether the lifestyle interventions based on mIoT could improve compliance and effectiveness of self-management in GDM women and to identify if it could improve the pregnancy outcome in Chinese GDM women.

Subjects and methods

Study design

This study employed a randomized and controlled trial design. Data were collected at baseline (pre-intervention) and before admission to hospital for delivery (post-intervention).

Sample calculation

Based on the formula for comparing the rates of two independent samples ($n_1 = n_2 = 1641.6 \left[\frac{(u_\alpha + u_\beta)}{\sin^{-1} \sqrt{p_1} - \sin^{-1} \sqrt{p_2}} \right]^2$), there was the sample size in each group to detect a 20% difference in the effective rate of blood glucose in the pretest with a power of 90% ($p_1 = 0.80$ for the mIoT-based lifestyle interventions, $p_2 = 0.60$ for the lifestyle interventions) (11,20). A one-sided significance level of 5% and a dropout rate of 20% was 120.

Ethical considerations

This study was approved by the Ethics Committee of the Hospital [No. 201701095]. Written informed consent was obtained from all participants. This study has been registered at the Chinese Clinical Trial Registry, and the trial registration number is [No. ChiCTR2000038031]. Patients were informed that their participation was voluntary and that they could withdraw from the study at any time. They were reassured that their withdrawal would not prevent them from receiving the care that they would normally receive. The participants were identified using research codes, and the research information remained confidential.

Subjects

A total of 240 GDM women were recruited for study participation at a tertiary hospital in southern China. Inclusion criteria: (1) women aged between 18 and 35 years (21); (2) women with a singleton pregnancy; (3) women with a diagnosis of GDM by the 75-g OGTT test at 24–28 weeks of gestation, 1-h postprandial

plasma glucose < 10 mmol/L, and 2-h postprandial plasma glucose < 8.5 mmol/L (2,3,22); (4) women who had to be capable of using the mobile application; and (5) women who were willing to provide informed consent. Exclusion criteria: (1) women with a diagnosis of preexisting diabetes; (2) women with chronic hypertension; (3) women with pre-pregnancy obesity; (4) women with poorly controlled hyperglycemia or hypoglycemia; (5) women with polycystic ovarian syndrome; (6) women with placenta previa; and (7) women with clinical exercise contraindications, such as threatened abortion.

Simple randomization was used to allocate the 240 participants to either the control or intervention group by a registered nurse with completed trial training before assisting in randomizing participants. All eligible women were labeled from 001–240 according to the hospitalization number from lowest to highest. Then, three digits ranging from 001–240 were produced by the random number table without duplication. Women whose numbers corresponded to the first 120 valid digits were assigned to the control group and the rest were assigned to the intervention group. This randomized group information was concealed in a sealed envelope. According to the order of hospitalization number, the test controller opened the corresponding envelope and assigned the woman to the corresponding group. This study did not use blind design. It is the visible and different management models, and the evaluation indicators of intervention effects are objective and measurable. Participants received no compensation for their participation in the study.

Intervention methods

All nutritionists and nutrition specialist nurses involved in the study would have completed this training with 10 years of work experience. The nutritionists and nutrition specialist nurses instructed the participants to do more physical activity, self-monitor blood glucose and improve diet quality based on medical nutrition therapy via a one-day outpatient visit. In addition, engineers and specialist nurses guided the participants in the intervention group to log on to the mIoT-based application and to use the application for real-time input of their nutrition, exercise, blood glucose, and weight. The nutritionists monitored and managed these participants remotely through the application and went on up to delivery. The participants in the control group did not use the mIoT-based blood glucose management application but followed the instructions of the nutritionists to keep a diary diet, exercise, and blood glucose level, and to manage abnormal blood glucose via WeChat up to delivery.

All instruments in this study were tested and showed adequate validity and reliability. Participants were randomly allocated into two groups. This study was conducted by obstetricians, nutritionists, specialist nurses, and software engineers, and they underwent standardized training to take homogeneous interventions.

Medical nutrition therapy

A 24-h dietary review was conducted to identify participants' dietary habits. Then, the obstetricians and nutritionists designed an appropriate and individualized diet for the participants according to their body mass index (BMI), desired body weight, physical activity, habits, and daily energy they needed (2,13,14). The dietary plans were based on both the principle of meal distribution and differential proportions of nutrients. The caloric intake was calculated based on pre-pregnancy BMI and desirable weight gain, including 35–40 kcal/kg desirable body weight for underweight women, 30–35 kcal/kg desirable body weight for normal-weight women, and 25–30 kcal/kg desirable body weight for overweight women (22).

After determining the daily energy intake in combination with labor intensity and gestational week, a calorie increase of 200 kcal per day was needed daily for both middle and late pregnancy with a basal calorie intake of 1800 kcal a day. The ratio of carbohydrate: protein: fat was 50:20:30. In addition, carbohydrates were not less than 175 g per day, food with a lower glycemic index was preferred (glycemic index < 55), and food with high fiber was preferred. The calories of the three meals should account for 10%–15%, 30%, and 30% of the daily total intake, and each extra meal could account for 5%–10% of the total energy. Three meals and three extra meals per day can help to reduce blood glucose fluctuations. Guidance on how to substitute foods with similar foods was also given to them, as was the recommendation to take in more than 12 types of foods per day and more than 25 types per week (2,12–14,22).

Physical activity

Participants were encouraged to perform moderate-intensity aerobic exercise lasting 30 min, 1 h after a meal, to achieve a target heart rate of 50%–60% of the age-predicted maximum heart rate at which the subject should be able to speak with relative ease. The total time for exercise was recommended to be more than 150 min per week (2,14,22,23).

Self-monitoring blood glucose and management

mIoT integrated maternal and fetal remote monitoring, early warning, and health guidance, with the purpose to enable gestational diabetes patients to obtain accurate and continuous nutrition guidance. The engineers developed a mIoT-based smart medical cell phone application. The specialist nurses instructed participants in the intervention group to continuously self-monitor blood glucose four times a day on at least 5 days a week and record their blood glucose levels in the mIoT-based application in a timely way. The monitoring times were while fasting once daily, post-prandial three times, and selectively 2 h post-prandial. The target blood glucose ranges were fasting blood glucose (FBG) < 5.3 mmol/L (95 mg/dl), 1-h postprandial < 7.8 mmol/L (140 mg/dl), and 2-h postprandial < 6.7 mmol/L (120 mg/dl) (2,13,14,21). When their blood glucose was assessed as well as controlled by the specialist nurse online, the participants changed to measuring 2–3 days per week at the same times. The specialist nurses also educated the participants on how to recognize the signals of low blood glucose and carry 15 g of simple carbohydrates, such as candies and sugary drinks, for treatment.

Remote management of nutrition and blood glucose based on mIoT

An engineer and a specialist nurse guided the participants in the intervention group to log on to the mIoT-based application and instructed them how to use the application to input their nutrition, exercise, blood glucose, and weight status in real-time. This application was applied for remotely monitoring and managing maternal and fetal nutrition and blood glucose level based on mIoT technology. The personal basic information, identity authentication, disease history, birth history, current pregnancy status, and related risk factors were collected by the application.

The participants in the intervention group recorded their daily dietary type and quantity, exercise, fingertip trace blood glucose, and body weight in this application. All data collected in the whole process were transmitted to the hospital system through its cloud data center. When data such as abnormal blood glucose levels and weight gain were detected, the system automatically alerted participants and healthcare workers. When the nutritionists received an abnormal signal, they immediately analyzed the abnormal signal and determined the trend of its change. Meanwhile, the medical staff gave feedback to the participants and guided them on what to do next.

The nutritionists analyzed all the updated information online daily, such as their diet, exercise records, blood glucose, and weight, to observe the changes and trends in these data. Then, combined with fetal growth and symptoms of discomfort in participants, the nutritionists concretely guided them to adjust their diet and exercise plans through the application. Based on this guidance, participants could adjust their diet and exercise. This systematic, professional analysis helped them understand how balanced their diet was and how effective their exercise was through timely feedback, so that they could autonomously manage their weight and blood glucose level. After one week of proper medical nutrition and exercise intervention, if the blood glucose still failed to meet the target above, the online nutritionists and nutrition specialist nurses would tell participants to return to the hospital for insulin treatment, as recommended by the guidelines (21).

The control group management

The nutritionists and nutrition specialist nurses instructed the participants in the control group to do more physical activity, self-monitor blood glucose and improve diet quality based on medical nutrition therapy via a one-day outpatient visit. The programs of medical nutrition and physical activity interventions in the control group, as well as self-monitoring of blood glucose, were the same as those in the intervention group. Unlike the intervention group, the control group did not use a mIoT-based blood glucose management application but followed the instructions of the nutritionists to manage diet and exercise, and recorded the blood glucose level in a diary. When the participants had problems in the management of their blood

glucose, they could upload their diaries and consult with the nutritionists through WeChat during the pregnancy period and up to delivery.

Participants in the control group who did not achieve the target blood glucose levels (FPG ≥ 5.3 mmol/L or 2hPG ≥ 6.7 mmol/L) after one week of lifestyle intervention were referred to a nutritionist and a specialized nurse for further evaluation. If the pregnant women with GDM continued to have uncontrolled blood glucose despite adhering to standardized dietary management, exercise, and blood glucose monitoring, they were treated with insulin therapy as recommended by the guidelines, similar to the intervention group.

Intervention compliance management

Participants came to the diabetes clinic every 4 weeks for nutritional assessment and weight management. If participants did not return to the hospital, they were deemed to have a missed visit. If they were delivered in other hospitals, they were deemed to be missing cases.

Data collection

Data were collected after all participants provided informed consent. Data were collected from the mIoT-based blood glucose management application, patient's diary, and medical records. Baseline data were collected immediately after group randomization. Four weeks after the intervention, data about treatment compliance and management effect were collected for a second time. The third data collection was carried out at week 8 after the intervention. Glucose metabolism and lipid metabolism indexes were collected pre-delivery at the admission. Pregnancy outcome indexes were collected during the fourth labor stage.

- (1) The demographic and clinical information of the patients were measured only at baseline. Demographic information included patient age, birth history, basic blood pressure, pre-gestational BMI, educational level, economic income, and sleep status. Body weight (fasting measurements in underwear), glucose-metabolism indexes, and lipid metabolism indexes were recorded with the same instruments before the intervention and upon admission before delivery.
- (2) Diabetes self-care management (DSCM) is "the exercise of self-care the actual performance of self-care actions by individuals to manage their diabetes" (24). DSCM has complex relationships with diabetes knowledge (knowledge of the disease, healthy eating, being active, blood glucose monitoring, and taking medication), self-care agency (capabilities to perform self-care activities), and self-efficacy (belief in personal capabilities to act and in the results of acting (25)). The evaluation of the effectiveness of GDM self-management included four aspects, performing standardized dietary control, regular exercise, blood sugar monitoring, and control of blood glucose and body weight in the normal range. In this study, GDM self-care management was evaluated by the nutritionist based on the data from mIoT-based blood glucose management application, patient's diary, and medical records. Referring to the FIGO guidelines, if pregnancy weight gain was within the recommended standard range, it was normal weight gain; otherwise, it was abnormal (22).
- (3) The incidence of complications that were associated with GDM outcomes of both mothers and neonates was recorded, including preeclampsia, preterm labor, macrosomia, asphyxia neonatorum, and neonatal hypoglycemia. We also recorded and assessed the baby's multidimensional information, including their weight, blood glucose level, and subcutaneous fat thickness. The subcutaneous fat thickness was tested by a LYC-3 skin fold measuring instrument produced by Liaoning Medical Equipment Factor (25). Subscapular angle (slightly lateral lower scapular angle, parallel to the lower scapular angle, parallel to the spine) was tested. The proportion of macrosomia and neonatal hypoglycemia was collected and analyzed.

Statistical analysis

Statistical analysis was carried out using SPSS software version 26.0. Demographic information was expressed as frequencies, percentages, and means $\pm$ standard deviation (SD). Chi-square tests, Mann-Whitney U rank

sum tests, and independent *t*-tests were used to compare the demographic and clinical data, as well as pregnancy outcomes between the two groups. $p < 0.05$ was considered as the significant differences.

Results

Comparison of metabolism indexes before the intervention

There were 120 GDM women in the intervention group, including 4 dropouts or losses to follow-up for delivery in the local hospital. There were 120 GDM women in the control group, including 11 dropouts or losses to follow-up for delivery in the local hospital. Therefore, there were 116 and 109 GDM women in the intervention group and control group, respectively. The demographic and illness-related characteristics associated with GDM were shown in Tables 1 and 2. There were no significant differences in the general statistics, glucose metabolism indexes, lipid metabolism indexes, or related biochemical indexes between the two groups before the intervention ($p > 0.05$). Moreover, the insulin usage rate in the control group was significantly higher than that in the intervention group (12.8% vs 4.3%, $p = 0.021$).

Status of health behavior and blood sugar level after the intervention

As shown in Table 1, after the interventions, compliance with dietary control, regular exercise, and blood glucose monitoring of the intervention group was significantly superior to those of the control group at weeks 4 and 8 ($p < 0.05$). The incidence of hyperglycemia in the intervention group was markedly lower than that in the control group at weeks 4 and 8 after the intervention and in the hospital before delivery ($p < 0.05$). The incidence of high weight gain in the intervention group was notably lower than that in the control group in the hospital before delivery ($p < 0.05$).

Table 1. Comparison of general data, status of behavior and blood sugar control of the 2 groups [N (%)].

Variables	Classification	Intervention group (n, 116)	Control group (n, 109)	χ^2, t, z	<i>p</i>
Before intervention					
Primipara, cases		78 (67.2)	71 (65.1)	0.111	0.739
Age		30.56 ± 3.21	30.29 ± 3.19	0.639	0.523
Pre-gestational BMI		22.01 ± 2.63	22.20 ± 2.93	-0.513	0.609
Pre-gestational BMI	Relatively slim	14 (12.1)	18 (16.5)	0.068	0.946
	Normal	84 (72.4)	72 (66.1)		
	Overweight	18 (15.5)	19 (17.4)		
	Obesity	5 (4.3)	9 (8.3)		
Economic Income	< 5000	25 (21.6)	17 (15.6)	0.762	0.446
	5,000–10,000	62 (53.4)	65 (59.6)		
	10,000–20,000	24 (20.7)	18 (16.5)		
	Above 20,000	5 (4.3)	9 (8.3)		
Education	Junior high school or below	16(13.8)	8(7.3)	0.647	0.518
	Senior high school or technical secondary school	54(46.6)	57(52.3)		
Sleep status	Undergraduate or above	46(39.7)	44(40.4)	0.546	0.585
	Good	52(44.8)	47(43.1)		
	Difficult to fall asleep	19(16.4)	28(25.7)		
	Easily awakened	30(25.9)	25(22.9)		
	Dreaminess	15(12.9)	9(8.3)		
Week 4 after the intervention					
Dietary control		98(84.5)	66(60.6)	16.288	< 0.001
Regular exercise		78(67.2)	54(49.5)	7.260	0.007
Regular blood sugar monitoring		89(76.7)	62(56.9)	10.025	0.002
Hyperglycemia		23(19.8)	38(34.9)	6.428	0.011
Week 8 after the intervention					
Dietary control		106(91.4)	72(66.1)	21.808	< 0.001
Regular exercise		97(83.6)	67(61.5)	13.956	< 0.001
Regular blood sugar monitoring		105(90.5)	78(71.6)	13.303	< 0.001
Hyperglycemia		9(7.8)	29(26.6)	14.221	< 0.001
Pre-delivery at the admission					
Hyperglycemia		5(4.3)	21 (19.3)	12.299	< 0.001
Use of insulin		5(4.3)	14(12.8)	5.293	0.021
Low weight ¹ gain		8(6.9)	14(12.8)	2.253	0.133
High weight gain		9(7.8)	20(18.3)	5.613	0.018

Table 2. Comparison of glycometabolism indexes, lipid metabolism indexes and related biochemical indexes of the two groups before the intervention (mean $\pm$ SD).

Variables	Intervention group (n = 116)	Control group (n = 109)	t	p
BMI	23.94 $\pm$ 2.69	24.04 $\pm$ 3.15	-0.268	0.789
Systolic pressure (mmHg)	113.10 $\pm$ 6.18	112.60 $\pm$ 7.03	0.575	0.566
Diastolic blood pressure (mmHg)	64.47 $\pm$ 4.96	63.49 $\pm$ 4.49	1.555	0.121
Fasting OGTT (mmol/L)	4.79 $\pm$ 0.37	4.83 $\pm$ 0.46	-0.610	0.543
OGTT 1 h (mmol/L)	9.99 $\pm$ 1.23	10.05 $\pm$ 1.30	-0.378	0.706
OGTT 2 h (mmol/L)	8.80 $\pm$ 1.38	8.87 $\pm$ 1.07	-0.391	0.696
FPG (mmol/L)	4.78 $\pm$ 0.61	4.74 $\pm$ 0.48	0.576	0.562
2 h PG (mmol/L)	6.74 $\pm$ 1.67	6.62 $\pm$ 1.58	0.575	0.566
FINS (μ U/mL)	7.36 $\pm$ 2.43	7.41 $\pm$ 2.36	-0.157	0.876
2 h INS (μ U/mL)	17.33 $\pm$ 9.98	17.73 $\pm$ 8.85	-0.320	0.749
Fasting C-peptide (ng/mL)	1.57 $\pm$ 0.44	1.54 $\pm$ 0.48	0.343	0.732
Postprandial 2 h C-peptide (ng/mL)	4.25 $\pm$ 2.04	4.09 $\pm$ 1.96	0.567	0.571
HbA1c (%)	5.04 $\pm$ 0.39	5.08 $\pm$ 0.36	-0.709	0.479
HOMA-IR	1.58 $\pm$ 0.61	1.57 $\pm$ 0.57	0.067	0.947
TG (mmol/L)	2.33 $\pm$ 1.03	2.21 $\pm$ 1.03	0.884	0.378
TCH (mmol/L)	5.74 $\pm$ 0.84	5.77 $\pm$ 0.86	-0.248	0.805
HDL-C (mmol/L)	1.91 $\pm$ 0.41	1.90 $\pm$ 0.28	0.359	0.720
LDL-C (mmol/L)	2.67 $\pm$ 0.63	2.73 $\pm$ 0.66	-0.684	0.494
CRP (mg/L)	0.82 $\pm$ 0.50	0.71 $\pm$ 0.60	1.522	0.130

BMI, Body mass index; OGTT, oral glucose tolerance test; FPG, fasting plasma glucose; 2 h PG, 2-hour postprandial plasma glucose; FINS, fasting insulin; 2 h INS, 2 hour post-meal insulin; HbA1c, Glycosylated hemoglobin; HOMA-IR, Homeostasis model assessment of insulin resistance; TG, Triglycerides; TCH, Total cholesterol; HDL-C, High density lipoprotein cholesterol; LDL-C, Low density lipoprotein cholesterol; CRP, C-reactive protein.

Comparison of glucose and lipid metabolism indexes after the intervention

As shown in Table 3, after the intervention, the BMI, diastolic blood pressure, FPG, and 2 h PG in the intervention group were all significantly lower than those in the control group, as were the levels of TG, TCH, LDL-C, and CRP ($p < 0.05$). The HDL-C levels in the intervention group were markedly higher than that in the control group ($p < 0.05$).

Analysis of the obstetric outcome indicators

After the intervention, the incidence of preeclampsia, preterm birth, neonatal hypoglycemia, and fetal macrosomia in the intervention group was significantly lower than that in the control group, as were the fetal weight and neonatal subscapular fat thickness ($p < 0.05$, Table 1, Table 3).

Discussion

Based on Orem's (2001) self-care model and Orem's nursing theory, this study aimed to determine whether the mIoT-based lifestyle interventions could improve the compliance and effectiveness of self-management in GDM women.

Our results showed that as intervention compliance improved over 4 and 8 weeks after the intervention, the GDM women in the intervention group receiving mIoT-based lifestyle interventions had better control of blood glucose and weight gain than the control group receiving routine intervention through diabetes outpatient visits. However, only 14 of 21 (66.7%) participants with blood glucose exceeded the normal standard in the control group chose to take insulin therapy. This may be explained by the fact that GDM women experience a long treatment course, a complex condition, unsatisfied appetite, and increased psychological burdens, with the development of the maternal and fetal remote monitoring software via mIoT. GDM women in the intervention group could record their daily dietary type and quantity, exercise, fingertip trace blood glucose, and body weight in the application. All data collected in the whole process were transmitted to the hospital system through its cloud data center. The nutritionists analyzed all the updated information online daily. When data such as an abnormal blood glucose level and weight gain were detected, the application automatically alerted the pregnant women and health care workers, and then the nutritionists would analyze the abnormal signals, determine the trend of its change, give feedback to the

Table 3. Comparison of glycol-metabolism indexes, lipid metabolism indexes, related biochemical indexes and pregnancy outcomes between the two groups after the intervention pre-delivery at the admission.

Variables	Intervention group (n = 116)	Control group (n = 109)	χ^2, t	p
BMI	25.34 ± 2.54	26.03 ± 3.10	-3.441	0.001
Systolic pressure (mmHg)	114.55 ± 7.56	116.08 ± 12.45	-1.463	0.145
Diastolic blood pressure (mmHg)	64.94 ± 8.0	71.48 ± 9.05	-7.939	< 0.001
FPG (mmol/L)	4.48 ± 0.50	4.69 ± 0.49	-2.727	0.007
2 h PG (mmol/L)	5.69 ± 0.63	6.49 ± 1.22	-3.609	< 0.001
FINS (μU/mL)	7.52 ± 1.74	7.36 ± 2.27	0.531	0.596
2 h INS (μU/mL)	15.57 ± 4.32	17.24 ± 6.22	-0.928	0.354
Fasting C-peptide (ng/mL)	1.80 ± 0.49	1.68 ± 0.91	0.887	0.376
Postprandial 2 h C-peptide (ng/mL)	4.04 ± 1.81	4.31 ± 2.43	-1.355	0.177
HbA1c (%)	4.85 ± 0.28	4.93 ± 0.30	-0.828	0.409
HOMA-IR	1.39 ± 0.47	1.54 ± 0.50	-1.648	0.101
TG (mmol/L)	1.58 ± 0.62	2.17 ± 0.93	-4.489	< 0.001
TCH (mmol/L)	5.26 ± 0.95	5.68 ± 1.23	-2.173	0.031
HDL-C (mmol/L)	1.87 ± 0.28	1.70 ± 0.33	2.555	0.011
LDL-C (mmol/L)	2.51 ± 0.79	2.82 ± 0.87	-2.014	0.045
CRP (mg/L)	0.81 ± 0.62	1.16 ± 0.61	-4.237	< 0.001
Pregnancy outcomes indexes after the intervention				
Preeclampsia	1(0.9%)	8(7.3%)	/	0.016
Preterm birth	5(4.3%)	13(11.9%)	4.429	0.035
Macrosomia	0(0%)	5(4.6%)	/	0.025
Asphyxia Neonatorum	1(0.9%)	6(5.5%)	/	0.059
Neonatal Hypoglycemia	3(2.6%)	15(13.8%)	9.535	0.002
Gestational age at birth (w)	39.02 ± 1.18	38.30 ± 1.59	3.560	< 0.001
Fetal weight (kg)	3.17 ± 0.29	3.32 ± 0.42	-3.034	0.003
Neonatal subscapular fat thickness (mm)	4.46 ± 1.09	4.85 ± 0.88	-2.913	0.004

BMI, body mass index; OGTT, oral glucose tolerance test; FPG, fasting plasma glucose; 2 h PG, 2-hour postprandial plasma glucose; FINS, Fasting insulin; 2 h INS, 2 hour post-meal insulin; HbA1c, Glycosylated hemoglobin; HOMA-IR, Homeostasis model assessment of insulin resistance; TG, Triglycerides; TCH, Total cholesterol; HDL-C, High density lipoprotein cholesterol; LDL-C, Low density lipoprotein cholesterol; CRP, C-reactive protein.

pregnant women, and guided them on what to do next. This systematic, professional analysis and guidance helped pregnant women understand how balanced their diet was and how effective their exercise was. So that the mIoT-based lifestyle interventions could improve compliance and effectiveness of self-management in GDM women.

This study also showed that the FPG and 2 h PG levels of the intervention group were significantly lower than those in the control group. The FPG and postprandial blood glucose in the intervention group were all improved than those in the control group. The main factors acting against insulin during pregnancy are a series of hormones secreted by the placenta, mainly human placental prolactin, progesterone, prolactin, estrogen, and cortisol (25). As pregnancy progresses, the production of these hormones also gradually increases, resulting in decreased sensitivity of the surrounding tissues to the insulin response and an increase in the anti-insulin effect (26). The insulin resistance of GDM women is more significant (26). Individualized exercise therapy can help improve insulin resistance (23). Importantly, in this study, mIoT-based remote management was available to accurately record the diet and exercise of the GDM women. It enabled nutritionists to provide individual guidance and adjustment according to the indicators recorded. Therefore, the mIoT-based lifestyle interventions could improve compliance of self-management in GDM women and then improve their glucose metabolism and thereby long-term weight control outcomes.

However, despite some improvement in FPG and 2 h PG, no significant change in HbA1c was observed in this study. The non-significant improvement in HbA1c may be attributed to the relatively short intervention period (12 weeks) since it reflects the average blood glucose level over the past 2–3 months (27). The rapid improvement in FPG and 2hPG suggests that mIoT is more sensitive to immediate blood glucose regulation, whereas changes in HbA1c require longer-term intervention (such as ≥20 weeks) (28). This provides direction for future research design.

The levels of TG, TCH, LDL/C, and CRP in the intervention group were significantly lower than those in the control group, and the incidences of preeclampsia, premature delivery, macrosomia, and neonatal hypoglycemia in GDM women were significantly lower than that in the control group. The weight and fat thickness of the subscapular angle of newborns in the intervention group were significantly thinner than those in the control group. GDM is an inflammatory disease that can destroy islet cells and aggravate insulin resistance (4). Plasma CRP is a sensitive indicator of

chronic subclinical inflammation and a predictor of insulin resistance syndrome that is independently associated with insulin sensitivity and metabolic disorders (4). Studies have shown that for every 1 mmol/L increase in FBG levels in GDM women, the risk of preterm delivery increased by 23% (29). Dyslipidemia is closely related to adverse pregnancy outcomes in late pregnancy (30). For instance, maternal triglycerides as a risk factor for pre-eclampsia (31). Abnormally elevated TG is closely related to premature delivery, macrosomia, and poor blood glucose control (32). The blood glucose in the serum of GDM women can enter the fetus through the placenta, hyperglycemia can stimulate the fetus to produce a large amount of insulin and result in progressive glucocorticoid anatomization. Therefore, increasing protein and fat synthesis could result in larger children and newborn hypoglycemia (8). This study suggested that mIoT-based lifestyle interventions could help GDM women improve their glucose metabolism and lipid metabolism by improvement of diet and exercise self-management compliance. Finally, pregnancy outcomes, such as preeclampsia, premature delivery, macrosomia, and neonatal hypoglycemia, could be improved by mIoT-based lifestyle interventions. The ultimate benefit of health education and management is behavior change (33). Based on Orem's self-care model and nursing theory, health education and management using a mIoT platform may be an effective method to improve health outcomes and can be applied to a wide range of people.

This study also has some limitations. First, the elderly pregnant women aged over 35 years old didn't join in our study. According to the "Expert Consensus on Pregnancy Management in Advanced Maternal Age," advanced maternal age (≥ 35 years) itself constitutes an independent risk factor, with significantly higher GDM occurrence rate, degree of insulin resistance, and risk of complications compared to pregnancies at optimal reproductive age (34). Second, this study did not use a blind design for the visible and different management models. Third, in this study, assessing DM self-care management was according to the GDM guidelines by nutritionists, not the Diabetes Self-Management Scale (DSMS). Finally, the risk factors for developing complications could not be further analyzed with the small sample size. Although dietitian-based assessments can capture GDM-specific behaviors, future research needs to integrate the DSMS scale to enhance the comparability of the two methods and explore the application effectiveness of the DSMS scale in different regions. The current intervention method did not evaluate the role of self-efficacy (such as emotional management) among GDM pregnant women, but its objective impact on core outcome indicators (blood glucose and body weight) was limited.

Conclusion

In conclusion, it suggested that lifestyle interventions using a mIoT platform based on Orem's self-care model and nursing theory have a positive clinical effect on the self-management of GDM pregnant women in China. GDM or DM pregnant women with the use of mIoT-based interventions may utilize drug strategies and self-management more effectively over long periods. However, multicenter research should be performed to verify the effect of the mIoT-based integrated lifestyle intervention model. Future studies with large sample sizes and long follow-ups are suggested to further examine the effects of mIoT-based integrated lifestyle interventions on GDM women.

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JW drafted this article and was the principal investigator for the project. LZ, HH, WZ, QZ, and XL supervised the data collection. LZ helped draft the manuscript. YW and BL provided support for baseline data analysts and YZ and QZ provided support for the pre and posttest data analysts. All authors read and approved the final version of the manuscript.

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Author contributions

CRedit: Jiying Wen: Conceptualization; Liping Zhou: Conceptualization; Haibin Hu: Conceptualization; Dongmei Duan: Conceptualization; Wenxia Zou: Conceptualization; Yuheng Zhou: Conceptualization; Xiaohong Lin: Conceptualization; Yuntao Wu: Conceptualization; Qimeng Zhao: Conceptualization; Qingguo Zhao: Conceptualization; Bing Li: Conceptualization.

Article highlights

1. For pregnant women with GDM, this visual IoT technology facilitates continuous self-monitoring of their diet, blood glucose, physical activity, and body weight, enabling them to master appropriate glycemic management methods and thereby enhancing their self-management confidence and adherence.
2. For healthcare providers, this technology supports systematic tracking and analysis of trends in blood glucose and body weight among GDM patients, allowing real-time and effective adjustments in combination with dietary and physical activity data.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Ethical approval

This study was approved by the Ethics Committee of Guangdong Women and Children Hospital [No. 201701095].

Data availability statement

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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